When Does Mean-Field Structure Matter? Distribution Dependence, Representation Collapse, and Architectural Constraints in Deep BSDE Market-Making

Christian Garry
Department of Mathematical Sciences
Durham University
christian.garry@durham.ac.uk

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Abstract

We investigate when and why distribution-dependent mean-field coupling becomes learnable in deep BSDE solvers for optimal market-making. Starting from the Cont–Xiong (2024) dealer market model, we implement four law encoders (moments, quantiles, histogram, DeepSets) to represent the population distribution, and test whether the learned optimal control depends on population shape. We uncover three failure modes that can mask genuine mean-field effects: (i) a broken coupling pathway where the BSDE generator does not use the law embedding; (ii) BatchNorm systematically erasing broadcast (shared) features that are constant across the training batch; and (iii) mean-pooled permutation-invariant encoders (DeepSets) failing to capture distributional variance under symmetric inputs. After correcting all three, we show that the competitive factor $h(\Phi(\mu_t))$ varies by $4\times$ across population shapes, optimal quotes shift by $2\times$, and quote skew flips sign — demonstrating genuine distribution-dependent optimal control. The effect is robust across encoder types (MomentEncoder and QuantileEncoder produce qualitatively identical results) and penalty regimes ($\phi = 0.01$ to 0.5). We extend the model with adverse selection (price-dependent execution), breaking the dimensional reduction that made the base problem effectively one-dimensional, and map the stability frontier across penalty strength, adverse selection intensity, and coupling type. The central finding is that *representation determines learnability*: mean-field effects are present in the model but invisible to the solver unless the law encoder exposes the relevant distributional features.

Keywords: mean-field games, backward stochastic differential equations, deep learning, market-making, law encoders, BatchNorm, representation collapse

1 Introduction

Mean-field game (MFG) and McKean–Vlasov (MV) formulations promise to capture the aggregate effects of interacting agents in large-population stochastic control problems. In market-making, this means each dealer's optimal quotes should depend not just on their own inventory but on the distribution of inventories and quotes across the population Cont and Xiong [2024], Carmona and Delarue [2018].

Deep BSDE methods Han et al. [2018], E et al. [2017] offer a scalable computational framework for such problems, and Han, Hu, and Long Han et al. [2022] extended them to MV-FBSDEs via fictitious play. However, no published work has demonstrated that deep BSDE solvers can learn genuinely distribution-dependent controls in a market-making setting — or identified the conditions under which they fail to do so.

In a companion preprint Garry [2026a], we showed that a deep jump-BSDE solver achieves $< 0.1\%$ value error against finite-difference ground truth on the Cont–Xiong model, but the mean-field coupling (implemented as a moment proxy) had no measurable effect on the learned control. We attributed this to the weak-coupling regime ($\phi = 0.01$).

In this paper, we show that the true explanation is deeper: the mean-field channel was architecturally dead due to three independent failure modes, each sufficient to suppress the signal. After correcting all three, we demonstrate that:

1. Distribution-dependent competitive effects produce $4\times$ variation in fill rates and $2\times$ variation in optimal quotes across population shapes.
2. The effect is robust across handcrafted law encoders (moments, quantiles) but invisible to mean-pooled DeepSets under symmetric inputs.
3. Adverse selection (price-dependent execution) breaks the dimensional reduction $V = x + qs + \phi(t, q)$, making the problem genuinely multi-dimensional.
4. The stability frontier shifts when mean-field coupling is active.

Contribution. This is an empirical computational study. We do not prove convergence theorems. Our contribution is diagnostic: identifying *when and why* mean-field structure becomes learnable, and demonstrating that representation choice is the critical bottleneck.

2 Model and Formulation

We work with the Cont–Xiong dealer market model Cont and Xiong [2024] as described in Garry [2026a]. The key ingredients are:

- Mid-price $S_t = S_0 + \sigma W_t$ (driftless Brownian).
- Inventory q_t evolving by Poisson jumps (or diffusion approximation).
- Execution probability $f(\delta, \mu_t) = e^{-\alpha\delta} \cdot h(\mu_t)$, where h captures population competition.
- Objective: maximise discounted spread profits minus inventory penalty $\psi(q)$.

2.1 Mean-Field Coupling via Law Encoders

In Garry [2026a], the competitive factor $h(\mu_t)$ was approximated by a moment proxy $\hat{h}(t, \bar{q}_t, \bar{\delta}_t)$ trained on mean inventory and mean spread. This produced no measurable effect.

We replace this with a *learned* competitive factor:

$$h(\mu_t) = g\big(\Phi(\mu_t)\big), \qquad \Phi(\mu_t) = \text{LawEncoder}(\{(S_i, q_i)\}_{i=1}^N), \tag{1}$$

where Φ maps the empirical population to a fixed-size embedding and $g : \mathbb{R}^d \to (0, 1]$ is a learned competitive factor network (sigmoid output). Four encoders are tested:

- **MomentEncoder**: mean, variance, skewness, mean$|q|$, max$|q|$, std $\to \mathbb{R}^6$.
- **QuantileEncoder**: mean + 5 quantiles $\to \mathbb{R}^6$.
- **HistogramEncoder**: soft Gaussian histogram $\to \mathbb{R}^{20}$.
- **DeepSetsEncoder**: $\Phi = \rho(\frac{1}{N}\sum_i \psi(x_i))$ with learned $\psi, \rho \to \mathbb{R}^{16}$.

2.2 Adverse Selection Extension

To break the dimensional reduction $V = x + qs + \phi(t, q)$ proved in Garry [2026a], we introduce price-dependent execution:

$$f_a(\delta, \text{signal}) = e^{-\alpha\delta}(1 + \eta \cdot \text{signal}), \qquad f_b(\delta, \text{signal}) = e^{-\alpha\delta}(1 - \eta \cdot \text{signal}), \tag{2}$$

where signal_t is an EMA of normalised price increments and η controls adverse selection strength. The state becomes $(S, q, \text{signal}) \in \mathbb{R}^3$.

3 Three Failure Modes of Mean-Field Coupling

We identified three independent architectural failures, each sufficient to suppress the mean-field signal.

3.1 Failure 1: Generator Bypass

The BSDE generator $f(t, x, y, z)$ computes execution profits and thus determines the training loss. In our initial implementation, f inherited from the base class and used the old scalar moment proxy $\hat{h}(t, \bar{q}, \bar{\delta})$ rather than the law embedding $\Phi(\mu_t)$. The law encoder entered the subnet inputs (affecting Z prediction) but not the generator (affecting the BSDE dynamics). Since the loss is determined by the terminal mismatch $|Y_T - g(X_T)|^2$ propagated through the generator, the optimiser had no gradient signal connecting Φ to the loss.

Fix. Override f to compute $h = g(\Phi(\mu_t))$ and use it in execution probabilities. Store the current law embedding via `set_current_law_embed()` before each f evaluation.

3.2 Failure 2: BatchNorm Erasure of Broadcast Features

The law embedding $\Phi(\mu_t)$ is identical for all agents in the batch at a given time step (it represents the population, not the individual). BatchNorm normalises across the batch dimension:

$$\text{BN}(x) = \frac{x - \mathbb{E}_{\text{batch}}[x]}{\sqrt{\text{Var}_{\text{batch}}(x) + \epsilon}}.$$

When x is constant across the batch, $\text{Var}_{\text{batch}} = 0$ and the output is zero (or near-zero from ϵ). This erasure persists through all subsequent layers because the law contribution remains constant-like after mixing.

Fix. Two-stream architecture: state features pass through BatchNorm, law features bypass it entirely. The streams are combined additively after separate processing.

3.3 Failure 3: DeepSets Representation Collapse

The DeepSets encoder computes $\Phi = \rho(\frac{1}{N}\sum_i \psi(x_i))$. Under random initialisation, ψ is approximately symmetric. For zero-mean populations, positive and negative inventory contributions cancel under mean pooling, collapsing variance information:

Encoder	Embedding distance (std 0.1 vs 5.0)	Cosine similarity
DeepSets	0.003	1.000
MomentEncoder	24.98	varies
QuantileEncoder	9.84	varies

The DeepSets encoder produces nearly identical embeddings for populations with radically different variance. This is structural, not a training failure: mean pooling of symmetric features erases higher-order distributional information.

Implication. Permutation-invariant encoders based on mean pooling may fail to capture distributional properties beyond the mean under symmetric inputs unless explicitly trained to detect them. This is a representation limitation, not a coupling failure.

4 Architecture

4.1 Two-Stream Mean-Field Subnet

To prevent BatchNorm suppression while maintaining training stability:

- **State stream**: BN $\to$ Dense $\to$ BN $\to$ ReLU $\to$ Dense (standard Deep BSDE architecture).
- **Law stream**: Dense $\to$ ReLU $\to$ Dense (no normalisation).
- **Output**: state_output + law_output (additive).

The law embedding survives all layers because it never passes through BatchNorm.

4.2 Competitive Factor Network

$g : \mathbb{R}^d \to (0.1, 1]$ maps the law embedding to a scalar competitive factor via Dense $\to$ ReLU $\to$ Dense $\to$ Sigmoid, with a floor of 0.1 to prevent degenerate zero-execution regimes.

5 Numerical Results

All experiments at $\phi = 0.1$ unless stated, trained on a single NVIDIA RTX 3090 GPU.

5.1 Distribution-Dependent Competition

After correcting all three failure modes, the competitive factor $h(\Phi(\mu_t))$ varies substantially with population shape:

Table 1: Competitive factor and controls at $q = 0$ (MomentEncoder, $\phi = 0.1$).

Population	h	Z^q	δ_a	δ_b	ν_a	ν_b
Narrow (std 0.1)	0.455	+0.025	0.696	0.638	0.160	0.175
Medium (std 1.0)	0.280	−0.016	0.648	0.686	0.106	0.100
Wide (std 3.0)	0.107	−0.285	0.334	0.999	0.065	0.024

The competitive factor varies by 4×. Quotes shift by 2×. Under wide populations, the market maker narrows the ask (aggressive selling) and widens the bid (defensive), consistent with fierce competition for executions.

5.2 Encoder Robustness

Both handcrafted encoders produce qualitatively identical results (3.6–4× variation). DeepSets produces constant h, confirming the representation collapse is the bottleneck, not the coupling.

Table 2: Competitive factor across encoders ($\phi = 0.1$).

Encoder	h(narrow)	h(medium)	h(wide)
MomentEncoder	0.442	0.331	0.109
QuantileEncoder	0.449	0.345	0.125
DeepSets	0.427	0.427	0.427

Table 3: MV effect across penalty regimes (MomentEncoder).

ϕ	Y_0	h(narrow)	h(wide)	Gap
0.01	0.109	0.353	0.100	0.253
0.05	0.103	0.453	0.100	0.353
0.10	0.063	0.479	0.100	0.379
0.50	−0.143	0.169	0.100	0.069

5.3 Penalty Regime Sweep

MV coupling is active at all tested ϕ. The earlier "no effect at $\phi = 0.01$" was entirely an artifact of the broken architecture. At $\phi = 0.5$, Y_0 goes negative and the h-gap shrinks, suggesting solver instability rather than reduced coupling.

5.4 Model Hierarchy

Table 4: Model hierarchy comparison ($\phi = 0.1$, MomentEncoder, 1000 iterations).

Model	State	Y_0	MV effect
Base (no coupling)	2D (S, q)	0.377	—
MV only (MomentEncoder)	2D (S, q)	0.052	−0.325
Adverse only (no coupling)	3D (S, q, sig)	0.380	—
Adverse + MV (MomentEncoder)	3D (S, q, sig)	0.086	−0.294

MV coupling reduces Y_0 by ~ 0.3 in both 2D and 3D settings. Adverse selection alone barely changes Y_0 (difference 0.003), consistent with its role in reshaping value structure rather than level. The MV effect is comparable with and without adverse selection, suggesting the competitive discount is the dominant mechanism at $\phi = 0.1$.

5.5 Stability Frontier

The stability frontier is dominated by the penalty coefficient ϕ, not the adverse selection strength η.

Table 5: Stability frontier: $\phi\times$ penalty type (300 iterations, quick sweep).

Penalty	$\phi = 0.01$	$\phi = 0.1$	$\phi = 1.0$	$\phi = 5.0$
Quadratic	✓ (0.16)	✓ (1.52)	× (15.5)	× (128)
Exponential	✓ (0.32)	✓ (2.17)	× (31.1)	× (77.7)

Numbers in parentheses are $\max|Z_t|$. The boundary lies between $\phi = 0.1$ (stable) and $\phi = 1.0$ (diverged). Adverse selection ($\eta = 0$ to 2.0) does not shift this boundary — all η values are stable

at $\phi \leq 0.1$ and diverge at $\phi \geq 1.0$. At $\phi = 0.1$, MV coupling with MomentEncoder reduces Y_0 from 0.381 to 0.141 (no coupling vs moments), while DeepSets gives $Y_0 = 0.115$, confirming the coupling is active even in the stability-sweep regime.

6 Discussion

Representation determines learnability. The central finding is that mean-field effects are *present in the model* but *invisible to the solver* unless the law encoder exposes the relevant distributional features. This has implications beyond this specific problem: any deep learning system that conditions on shared population statistics via broadcast features must account for normalisation layer interactions.

Architectural lessons. BatchNorm, while critical for Deep BSDE training stability Han et al. [2018], is fundamentally incompatible with broadcast conditioning signals. The two-stream architecture (BN on per-agent features, no BN on shared features) resolves this cleanly. FiLM-style modulation Perez et al. [2018] would be a natural alternative.

Economic interpretation. The learned competitive factor $h(\Phi(\mu_t))$ maps wider inventory distributions to lower fill rates, consistent with the intuition that dispersed populations contain more aggressive competitors. The resulting policy adaptations (narrower ask under competition, wider bid) are economically rational inventory management responses.

7 Conclusion

We have shown that distribution-dependent mean-field coupling is learnable in deep BSDE market-making solvers, but only when three architectural conditions are met: the law embedding enters the BSDE generator, broadcast features bypass BatchNorm, and the encoder captures distributional variance. When all three are satisfied, the solver learns economically meaningful competitive effects that materially change optimal quotes.

The deeper lesson is methodological: null results in mean-field deep learning may reflect architectural failures rather than genuine model invariance. Diagnosing the failure mode — coupling pathway, normalisation, or representation — is essential before concluding that population structure does not matter.

Limitations. The competitive factor network learns a monotone mapping (wider $\rightarrow$ lower h) but we have not verified that this corresponds to a Nash equilibrium fixed point. The adverse selection model uses a simple EMA signal. The experiments are at $d = 2$–3.

Future work. Fictitious play convergence verification; FiLM conditioning as alternative to two-stream; DeepSets with variance-sensitive features (q^2 pooling); common noise via deep signatures Hu et al. [2025].

Code. `https://github.com/cgarryZA/DeepMVBSDEJ`

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